

Name: A Rauf

Roll No.

Section: The logo is a green shield with a purple border. At the top is a circular emblem with a book and a laurel wreath. The shield is divided into two quadrants: the left one contains a red cross with a stethoscope, and the right one contains a blue caduceus. Below the shield is a purple banner with the text "GRACE COLLEGE" in white, and "of Pharmacy & Health Sciences" in smaller white text below it.

GRACE COLLEGE
of Pharmacy & Health Sciences

Subject : Pharmaceutics
Grace Collage of pharmacy
and health science Khampur

Sr.No List of Experiment

- 1- Introduction of label
- 2- Preparation of isotonic solution
(0.9% NaCl normal saline)
- 3- Preparation of buffer solution
(Phosphate buffer)
- 4- Prepare the pure crystal by
crystallization.
- 5- Demonstration of simple distillation.
- 6- Demonstration of prescription
- 7- Preparation of simple ointment
- 8- Preparation of Sulphur ointment
- 9- Preparation of Iodine Tincture
- 10- Preparation of ORS. -
- 11- Determination of specific gravity of
sample solution.



Experiment No 1:

* Introduction of label:-

Labeling is comprehensive information about the drug quantity.

• Quantity:

Quantity refers to the strength and physical form in which it is supplied and the rules for the its storage and handling.

• Efficiency:

Efficiency refers to medical condition for the therapeutic effects of use as well as the proper dosage form.

• Safety:

Safety refers to potential strength, side effects, contain indications and other consequences of use.

• Label:

Label means a display of written printed graphic matter upon the immediate container or outside containers or wrappers of drug package.

Laboratory Label

1- Labeling:

Labeling means display of written printed graphic matter upon the immediate container or outside container or wrappers of drug package.

2- Manufacturing:

Manufacturing in relation to a drug means all operations including processing, compounding, formulation, filling, packaging, finishing and labeling with a view at its storage, sale and distribution.

3- Manufacturing date:

Manufacturing date means the date stated on the label of drug at which it is manufactured.

4- Expiry date:

Expiry date means the date stated on the label of drug after which the drug is not expected to retain its chemical efficiency, safety and quality.

5- Batch numbers:

Batch number means the label of the drug that identify the batch and permits the production history of the batch including all stages of manufacturing and control.

6- Generic name:

Generic name means the non proprietary or official name of

a drug as approved by the official government.

Brand name:

It is the proprietor name of a drug of each company.

Precautions:-



Shake well before use.

Keep away from children.

Not to be taken.

For external use only.

Store in a cool and dry place

Inflammable

Manufacturing license.

Labeling uses:

Labeling benefits both industry and patient.

- ii- The safe use of medicine depends on accurate reading of the labeling.
- iii- The purpose of providing details and accurate information on pharmaceutical labeling is to reduce medication error.

Sana



Experiment No: 2

★ Preparation of isotonic solution (0.9% NaCl normal saline)

Chemicals:-

- 0.9% normal saline
- 5% dextrose solution

Apparatus:-

- Weighing balance
- Stirrer
- Measuring cylinder
- Beaker
- Sodium chloride
- Water



Experiment No: 2

* Preparation of isotonic solution (0.9% NaCl normal saline)

Chemicals:-

- 0.9% normal saline
- 5% dextrose solution

Apparatus:-

- Weighting balance
- Stirrer
- Measuring cylinder
- Beaker
- Sodium chloride
- Water

Procedure:

- Accurately weighed 0.9g of NaCl on weighing balance.
- Measured water in a measuring cylinder.

- Took NaCl in a beaker and 50ml water is added into it and stirred properly to make a solution.
- Then made up the final volume up to 100ml by adding water.
- Finally labeled the beaker.

Storage:-

- Store in a cool and dry place.
- Store in a well closed container.

Applications:

- Isotonic solution are used to balance the electrolytes.
- It is used to flush wounds.
- It is also used in nasal washes.
- It is used intravenously in patients who are in unconsciousness.

Qamra

Experiment No. 3

★ Preparation of buffer solution (Phosphate buffer)

Chemicals:-

- Sodium hydroxide
- Di. sodium monophosphate
- Monobasic potassium phosphate
- Sodium chloride
- Distilled water
- 1M NaOH
- 1M HCl

Apparatus:-

- Beaker
- Stirrer
- Measuring cylinder
- Weighing balance
- Volumetric flask



Experiment No : 3

Preparation of buffer solution (Phosphate buffer)

Chemicals:-

- Sodium hydroxide
- Di-sodium monophosphate
- Monobasic potassium phosphate
- Sodium chloride
- Distilled water
- 1 M NaOH 4g / 100ml
- 1 M HCl 36.5g / 100ml

Apparatus:-

- Beaker
- stirrer
- Measuring cylinder
- Weighing balance
- Volumetric flask

Procedure:-

Mixture A

- Weight accurately 27.22 g of monobasic

potassium phosphate.

- Dissolved in 100ml of distilled water
- Take 50 ml from above solution and pour it in 200ml volumetric flask.

Mixture B

- Weight accurately 8g of sodium hydroxide. Dissolve it in 100ml of water.
- Take 22.4ml from above mixture B and pour it in mixture A.
- Make the final volume upto 200ml with water.
- Check the pH of solution. it should be 6.8.
- Adjust the volume and pH with 1M NaOH or 1M HCl.

Applications:

- It is used to stabilize the product by maintaining pH constant.

Experiment No. 4

★ Prepare the pure crystal by crystallization.

Apparatus :-

- Beaker
- Funnel
- China dish
- Burner
- Tripod stand
- Filter paper
- Wire gauze



Chemicals:-

- Impure substance
- Copper sulphate
- Solvent

Experiment No. 4

* Prepare the pure crystal by crystallization.

Apparatus:-

- Beaker
- Funnel
- China dish
- Burner
- Tripod stand
- Filter paper
- Wire gauze



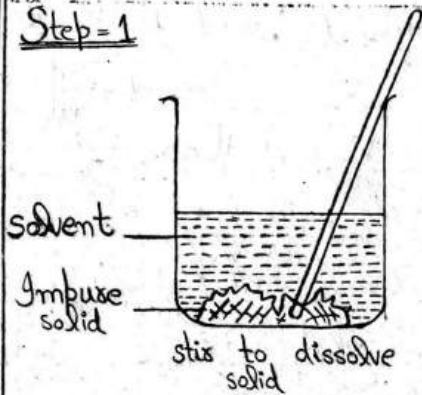
Chemicals:-

- Impure substance
- Copper sulphate
- Solvent

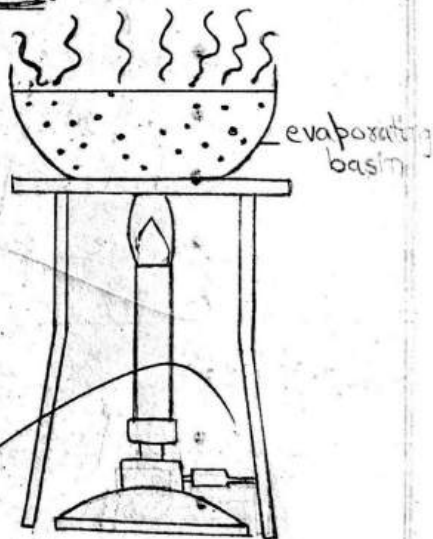
Procedure:-

- 1- Take 500ml of distilled water in a beaker and prepare saturated solution of copper sulphate.

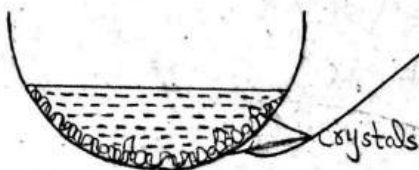
Step = 1



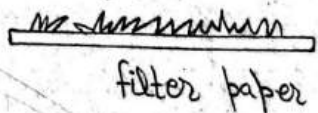
Step 2:-



Step 3:-



Step 4:-



2- Stir the solution with stirrer and filter through filter paper. On the filter paper impure residues are retained and in beaker we collect filtrate.

3- This filtrate consists of water and pure copper sulphate. Pour this filtrate in china dish and collect the crystal on a dry filter paper.

4- Allow crystal to dry and note their shape and colour.

Done



Experiment No : 5

Demonstration of simple distillation.

Apparatus:-

- Heating mantle
- Boiling flask with stirrer
- Thermometer
- Condenser
- Collecting flask

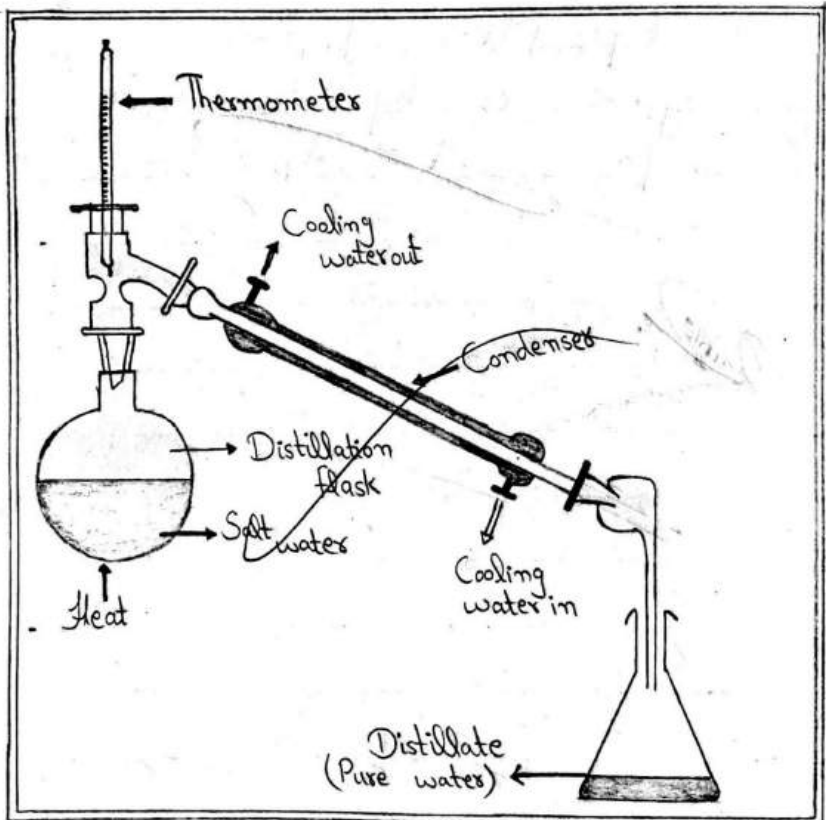


Fig:- Simple distillation

Experiment No: 5

Demonstration of simple distillation.

Apparatus:

- Heating mantle
- Boiling flask with stirrer
- Thermometer
- Condenser
- Collecting flask

Procedure:

- 1- Liquid is placed in boiling flask.
- 2- Heat is provided by heating mantle.
- 3- The liquid is boiled at its boiling point and then converted into vapours.
- 4- The condenser again converts the vapours into liquid by collecting them.
- 5- Finally pure liquid is collected in collecting flask leaving behind the solid impurity.

Applications:

1. It is used in the preparation of alcohol, ether and spirit.
2. It is used to extract volatile oil from crude drug.

Spring



Experiment No: 6

Demonstration of prescription.

Defination.

Prescription is an order written by a physician, dentist or any other registered medical practitioner to a pharmacist to compound and dispense a specific medication for the patient.

Parts of a prescription:-

A complete prescription should have the following parts:-

- Date
- Name, age, sex and address of the patient.
- Superscription
- Signature
- Inscription
- Prescriber's name
- Address, registration no. and signature



Date:-

Date must be written on the prescription by the prescriber at the same time when it is written.

Name, age, sex and address of patient:-

Name, age, sex and address of patient must be written on the prescription. If it is not written, then the pharmacist himself should ask the patient about these particulars and write down at the top of the prescription.

Superscription:-

The superscription is represented by a symbol Rx which is always written at the beginning of the prescription.

(Rx = you take)

Inscription:-

This is the main part of prescription it contains the names and quantities of the prescribed medicine or ingredients. The names of medicine or ingredients are written each on a separate line.

Subscription:-

This part of the prescription contains prescriber's directions to the pharmacist regarding the dosage form to be prepared and no. of doses to be dispensed it may also include instruction regarding the flavour of product, label etc.

Signature:-

It is usually abbreviated as sig on the prescription and consists of the directions to be given to

the patient.

These may include :

- The method of administration and application.
- The dose if the preparation is for internal dose
- The time of administration or application.
- The part of body where the preparation is to be applied, in case of external use.

The prescriber's name, address, registration no. and signature :-

The address and registration number may be printed but the prescriber's name must be hand-written and should be signed with ink.



Experiment No. 7

Preparation of simple ointment.

Apparatus:-

- Beaker
- Stirrer
- Weighing. balance
- Tripod stand
- Wire gauze
- Spirit lamp



Chemicals:-

- Wool fat 50g
- Hard paraffin 50g
- Cetosteryl alcohol 50g
- White and yellow wax 850g
- Simple ointment will be make 1000g

Experiment No: 7

Preparation of simple ointment.

Apparatus:-

- Beaker
- Stirrer
- Weighing balance
- Tripod stand
- Wire gauze
- Spirit lamp

Chemicals:-

- Wool fat 50g
- Hard paraffin 50g
- Cetosteryl alcohol 50g
- White and yellow wax 850g
- Simple ointment will be make 1000g.

Procedure:-

- 1- Mix the chemical with stirrer.
- 2- Heat all the chemicals until it

melts and form a homogenous mixture.

Uses:-

- 1- It is used as a base for preparation of medicated ointment.
- 2- It is only used for external purpose.

Sam



Experiment No: 8

Preparation of sulphur ointment.

Chemicals:-

- Precipitated sulphur 100g
- Mineral oil 100g
- White ointment 800g
- To make 1000g

Experiment No. 8

Preparation of sulphur ointment.

Definition:-

Sulphur ointment contains not less than 9.5 percent and not more than 10.5 percent of sulphur.

Chemicals:-

- Precipitated sulphur 100g
- Mineral oil 100g
- White ointment 800g
- To make 1000g

Procedure:-

- 1- Levigated the sulphur with mineral oil to form a smooth paste.
- 2- Then incorporated white ointment in it.

Packaging and storage:-

Preserve in well-closed containers, and avoid prolonged exposure to

excessive heat.

Uses:-

- 1- Sulphur is used to treat many kinds of skin disorders.
- 2- Sulphur ointment is used to treat scabies.

Sulphur



Experiment No : 9

★ Preparation of ORS.

Apparatus:-

- Beaker
- Stirrer
- Weighing balance

Chemicals:-

- NaCl 2.5g
- Sugar 30g
- Water 1000ml

Experiment No : 9

Preparation of ORS:-

ORS:-

ORS is the abbreviation of oral rehydration solution.

Apparatus:-

- Beaker
- Stirrer
- Weighing balance

Chemicals:-

- NaCl 2.5g
- Sugar 30g
- Water 1000 ml

Procedure:-

1. Weigh the chemicals accurately.
2. Add sugar and salt in a beaker.
3. Add small amount of water with constant stirring.

4 After mixing the chemical make up volume up to 1000ml with water.

Uses:-

It is used to balance electrolyte deficiency due to diarrhea and accidental fluid loss.

Same



Experiment No 10 :-

Preparation of iodine tincture.

Apparatus:-

- Beaker
- Stirrer
- Weighing balance

Chemicals:-

- Iodine 20g
- Sodium iodide 24g
- Alcohol 500 ml.
- Water q.s make alcohol

Experiment No: 10

Preparation of iodine tincture.

Definition:-

Tincture are alcoholic preparations containing active ingredients.

Apparatus:-

- Beaker
- Stirrer
- Weighing balance

Chemicals:-

- Iodine 20g
- Sodium iodide 24g
- Alcohol 500ml
- Water q.s to make 1000ml.

Procedure:-

1. Weigh all the chemicals accurately.
2. Add 20g iodine and 24g sodium iodide to 500ml alcohol.

3. Stirrer it and add purified water to make final volume upto 1000 ml.

Uses:-

- It is used as antiseptic.

Experiment No 11:

Determination of specific gravity of sample solution.

Apparatus:-

- Weighing balance
- Pycnometer

Chemicals:-

- Water
- Sample solution



Experiment No 11:-

Determination of specific gravity of sample solution.

Specific gravity:-

The specific gravity is the ratio between the density of a sample solution and a reference substance.

Unit:-

It has no unit.

Density:-

We measure density in mass per unit volume.

Units:-

- Gram per milliliter (g/ml)
- Grams per cubic centimeter (g/cm^3)
- Kilograms per liter (kg/L)

Pycnometer :



Apparatus:-

- Weighing balance
- Pycnometer

Chemicals:-

- Water
- Sample solution

Procedure:-

- 1- Wash pycnometer thoroughly.
- 2- Weight an empty pycnometer on weighing balance.
- 3- Add water in pycnometer and weight on weighing balance.
- 4- Subtract both the values.
- 5- Calculate the density of water with the help of formula.
- 6- Now add sample solution in pycnometer and weight it.
- 7- Calculate the density of sample.

8- Calculate the specific gravity of sample with the help of given formula:-

$$\text{Specific gravity} = \frac{\text{Density of sample}}{\text{Density of water}}$$

Same

